Statistics – 3

Population vs Sample Variance · Bessel's Correction · Percentile · Quartiles & IQR

04 April 2026 · Statistics – 3 · Lecture Notes

Variance (n vs n-1) | Bessel's Correction | Degrees of Freedom | Percentile | Quartiles & IQR

RECAP FROM LAST CLASS

Mean, Median, Mode → Measurement of **Central Tendency**

Variance, Standard Deviation → Measurement of **Dispersion**

Variance (σ^2) = $\Sigma(x_i - \mu)^2 / n$ (population) | SD (σ) = $\sqrt{\text{Variance}}$

§ 01 Population Variance vs Sample Variance (n vs n–1)

TWO FORMULAS FOR VARIANCE

Population variance uses **÷ n** → μ is the true population mean (fixed, external)

Sample variance uses **÷ (n–1)** → $\bar{x}$ is estimated from the sample (underestimates μ)

Population mean μ: not something you derive from data — it's a **fixed property of the population**.

$$\text{Population Variance: } \sigma^2 = \Sigma(x_i - \mu)^2 / n$$

$$\text{Sample Variance: } s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

$$\text{Population SD: } \sigma = \sqrt{(\Sigma(x_i - \mu)^2/n)} \quad \text{Sample SD: } s = \sqrt{(\Sigma(x_i - \bar{x})^2/(n-1))}$$

Why is sample mean $\bar{x}$ < population mean μ?

THE BIAS PROBLEM

Population = [1, 2, 3, ... 100] → true mean μ = 50.5

Sample (30 values) → mean x■ → tends to **underestimate** μ (x■ < μ)

Why? Sample selects values non-uniformly. Small samples often miss extremes.

Example [1...10], sample [7,8,9]: x■=8 > μ=5.5 | sample [1,2,3]: x■=2 < μ=5.5

→ On average across many samples, x■ < μ → this is called **bias**

→ Dividing by n makes variance even more underestimated → **biased estimator**

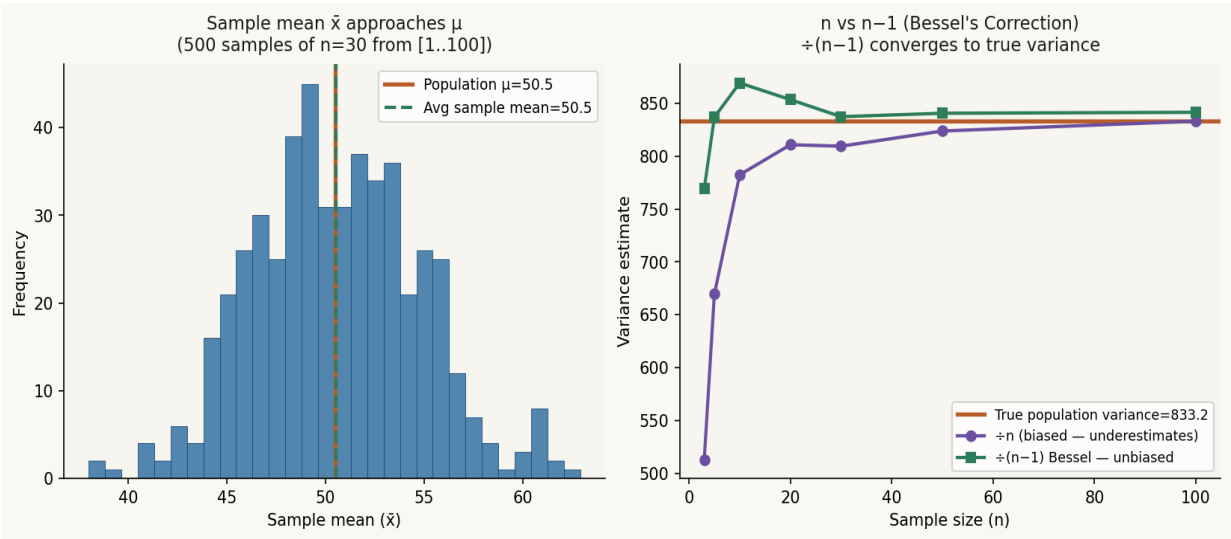

Figure 1: Sample means approach μ (left) | n vs n–1: Bessel's correction converges to true variance (right)

§ 02 Bessel's Correction — Why Divide by (n–1)?

BESSEL'S CORRECTION

Dividing by **(n–1) instead of n** inflates the result just enough to **cancel out the underestimate**.

1/n → makes variance smaller (biased) | 1/(n–1) → inflates → unbiased estimate

Result: $s^2 \approx\approx \sigma^2$ (sample variance approximates population variance)

This is why Python: np.var(data, ddof=1) for sample, ddof=0 for population.

Simulation Intuition:

Population [1..100], take 1000 bags of 5 marbles each → measure variance of each bag.

With ÷n: average of all bag variances < true variance (underestimates).

With ÷(n–1): average of all bag variances ≈ true variance (correct estimate).

§ 03 Degrees of Freedom

INTUITION

Population mean μ = **external** (fixed, not derived from sample data).

With n=5 values and known μ=163: x■+x■+x■+x■+x■ = n×μ = 5×163 = 815

If x■=200, x■=100, x■=300, x■=100 (4 free values, sum=700)

→ x■ is **forced**: 815 – 700 = 115. x■ has NO freedom.

→ Only **n–1 = 4 values are free** → df = n–1

```
Degrees of Freedom = n – 1
For sample: n–1 values are free to vary; the last is determined by the mean constraint
Mathematical fact: Σ(x■ – x■) = 0 always (sum of deviations from mean = 0)
```

Worked Example — Why Σ(x■ – x■) = 0:

x■	x■	x■ – x■	(x■–x■)²
39	47	39–47 = –13	169
47	47	47–47 = 0	0
52	47	52–47 = +5	25
61	47	61–47 = +14	196
41	47	41–47 = –6	36

Sum		–13+0+5+14+(–6) = **0**	426

TWO FACTS ABOUT N–1

Fact 1 (Mechanical): Only n–1 values are truly free in a sample. The last is determined by the mean constraint.

Fact 2 (Bias correction): Dividing by n–1 instead of n corrects the systematic underestimation of population variance.

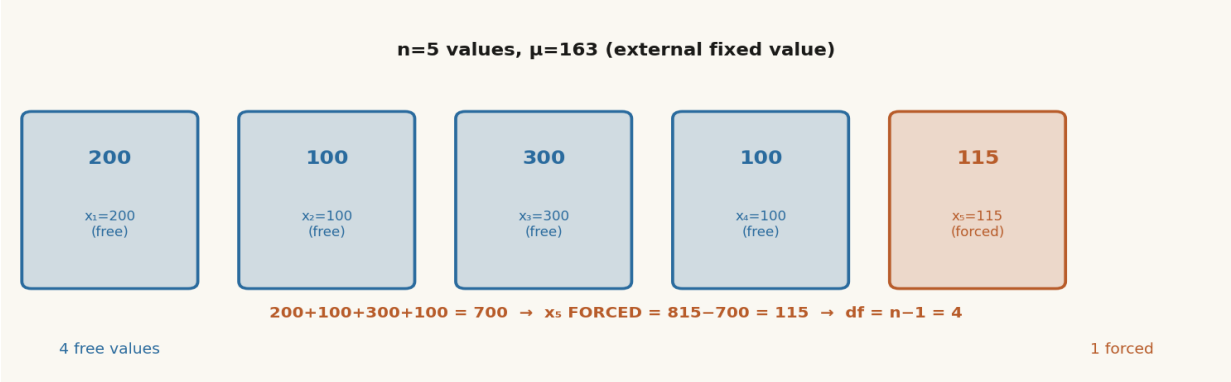

Figure 2: Degrees of freedom — 4 free values, x■ is forced once μ=163 is fixed

§ 04 Percentage & Percentile

Term	Definition	Formula	Example
Percentage	Ratio expressed out of 100 — what fraction of total?	% = (part/total) × 100 Can go above 100%	50/100 = 50% NOT > 100% for proportions
Percentile	What % of the group scored BELOW you?	Two formulas below	75th %ile → 75% scored less than you CAT → 96th %ile, 99th %ile

PERCENTILE INTUITION

Top exam → 100th percentile → 50/50 people → scored above all

→ one person → 50/100 → top → 100%

→ second → 49.9/100

75th percentile → 75% of people scored less than you.

40/200 and 15/200 (in context of class size)

Formula 1 — Nearest Rank (School method, jumps)

```
L = ■ (p/100) × n ■ (ceiling function — round UP)
p = desired percentile | n = total data points | L = rank/index
Result = value at position L in sorted data
```

```
Data (sorted, n=10): 12, 18, 24, 30, 35, 41, 47, 55, 63, 70
```

Percentile p	L = ■(p/100)×10■	Value at position L	Answer
p=30	L = ■0.3×10■ = ■3■ = 3	3rd value = 24	24
p=32	L = ■0.32×10■= ■3.2■ = 4	4th value = 30	30 (jumps!)
p=39	L = ■0.39×10■= ■3.9■ = 4	4th value = 30	30

Formula 2 — Linear Interpolation (most common, continuous scale)

```
L = (p/100) × (n – 1) + 1 (continuous scale, smooth)
If L is fractional → interpolate between adjacent values:
Result = value[floor(L)] + (fractional part) × (value[ceil(L)] –
value[floor(L)])
```

Example p=38, data=[12,18,24,30,35,41,47,55,63,70], n=10:

Step	Calculation	Value

L = (38/100)×(10–1)+1	0.38×9+1 = 3.42+1	L = 4.42 — wait, BODMAS:
Correct: L = (30/100)×(9)+1	0.3×9+1 = 2.7+1	L = 3.7 (for p=30)
rank 3 = 24, rank 4 = 30	fractional part = 0.7	
Result	24 + 0.7 × (30–24) = 24 + 4.2	= 28.2

Extended example p=58, data=[29,32,48,49,50,58,62,72,90,100], n=10:

Step	Calculation	Result
L = (58/100)×(9)+1	0.58×9+1 = 5.22+1	L = 6.22
Floor(L)=6 → value=58	Ceil(L)=7 → value=62	gap = 62–58 = 4
0.22 × 4	= 0.88	
Result	58 + 0.88	= 58.88 (58th percentile value)

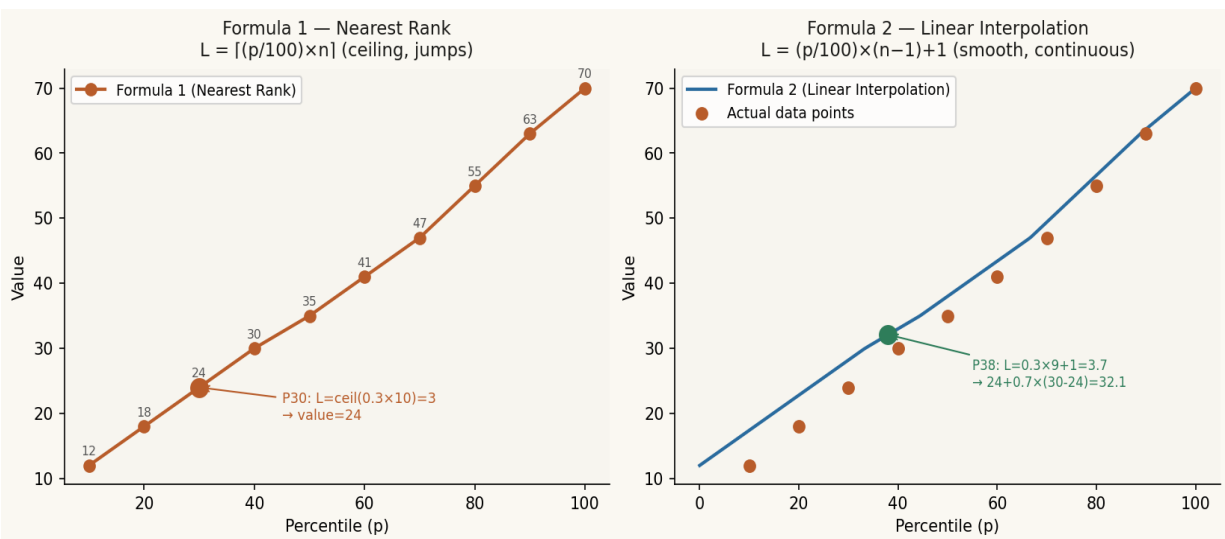

Figure 3: Formula 1 (nearest rank, jumps) vs Formula 2 (linear interpolation, smooth) — both on sorted data [12,18,24,30,35,41,47,55,63,70]

§ 05 Quartiles & IQR (+ Box Plot)

QUARTILES

Quartiles = special percentiles that split data into **4 equal parts**.

Q1 = 25th percentile (25% of data below) | Min → Q1 → Q2 → Q3 → Max

Q2 = 50th percentile (median)

Q3 = 75th percentile (75% of data below)

IQR (Interquartile Range) = Q3 – Q1 → place/region where **bulk of data lives**

```
IQR = Q3 – Q1 (middle 50% of data)
IQR covers 50% of data: from 25th to 75th percentile
SD = spread around mean | IQR = spread of middle 50% (robust to outliers)
```

IQR vs SD — Why IQR?

	SD	IQR
What it measures	Spread around mean	Spread of middle 50%
Affected by outliers?	Yes — sensitive	No — robust
Use case	Normal/clean data	Skewed/outlier-heavy data
Boxplot shows	Whiskers	The box itself

Box Plot (Data Visualization):

BOX PLOT STRUCTURE

[Min — Q1 |■ Q2 ■| Q3 — Max] with outliers shown as dots

- The **box** = IQR (Q1 to Q3)
- The **line inside box** = median (Q2)
- **Whiskers** extend to Min/Max within fences
- **Dots beyond whiskers** = outliers

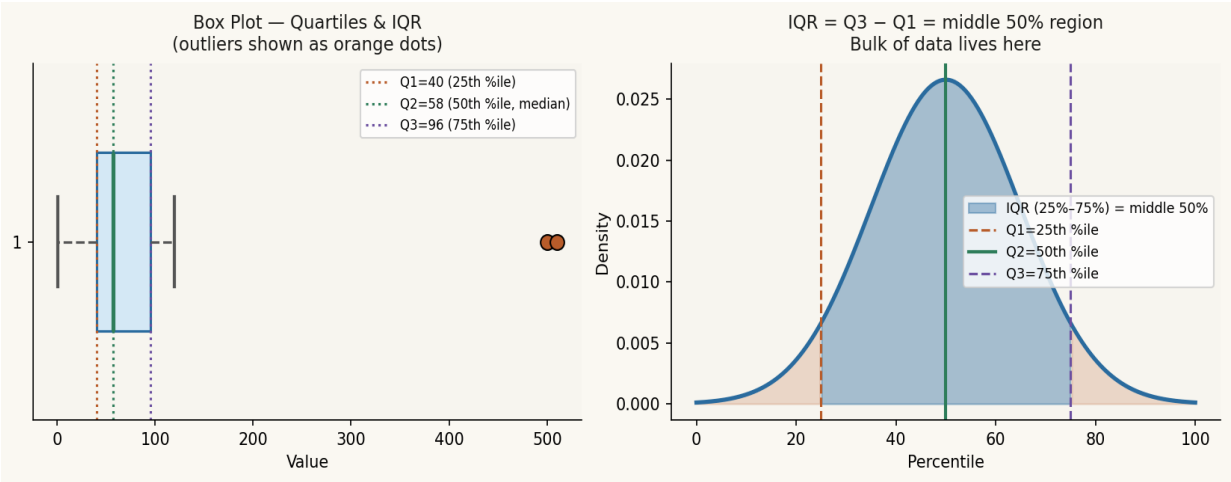

Figure 4: Box plot showing Q1/Q2/Q3 with outliers (left) and IQR as middle 50% region (right)

§ 06 Summary — All Key Formulas

Concept	Formula / Rule	Key Point
Population Variance	$\sigma^2 = \Sigma(x_i - \mu)^2 / n$	μ is external/fixed
Sample Variance	$s^2 = \Sigma(x_i - \bar{x})^2 / (n-1)$	n–1 = Bessel's correction
Bessel's Correction	Divide by (n–1) not n	Cancels underestimation bias
Degrees of Freedom	df = n–1	Only n–1 values are free
Σ deviations	$\Sigma(x_i - \bar{x}) = 0$	Always true — mathematical fact
Percentage	% = (part/total)×100	What fraction of total?
Percentile	% of group scored below you	75th %ile: 75% scored less
Nearest Rank	$L = \lceil (p/100) \times n \rceil$	School method, jumps
Linear Interp.	L = (p/100)×(n–1)+1	Most common, smooth/continuous

Quartiles	Q1=25th, Q2=50th, Q3=75th %ile	Splits data into 4 parts
IQR	IQR = Q3 – Q1	Middle 50%, robust to outliers

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